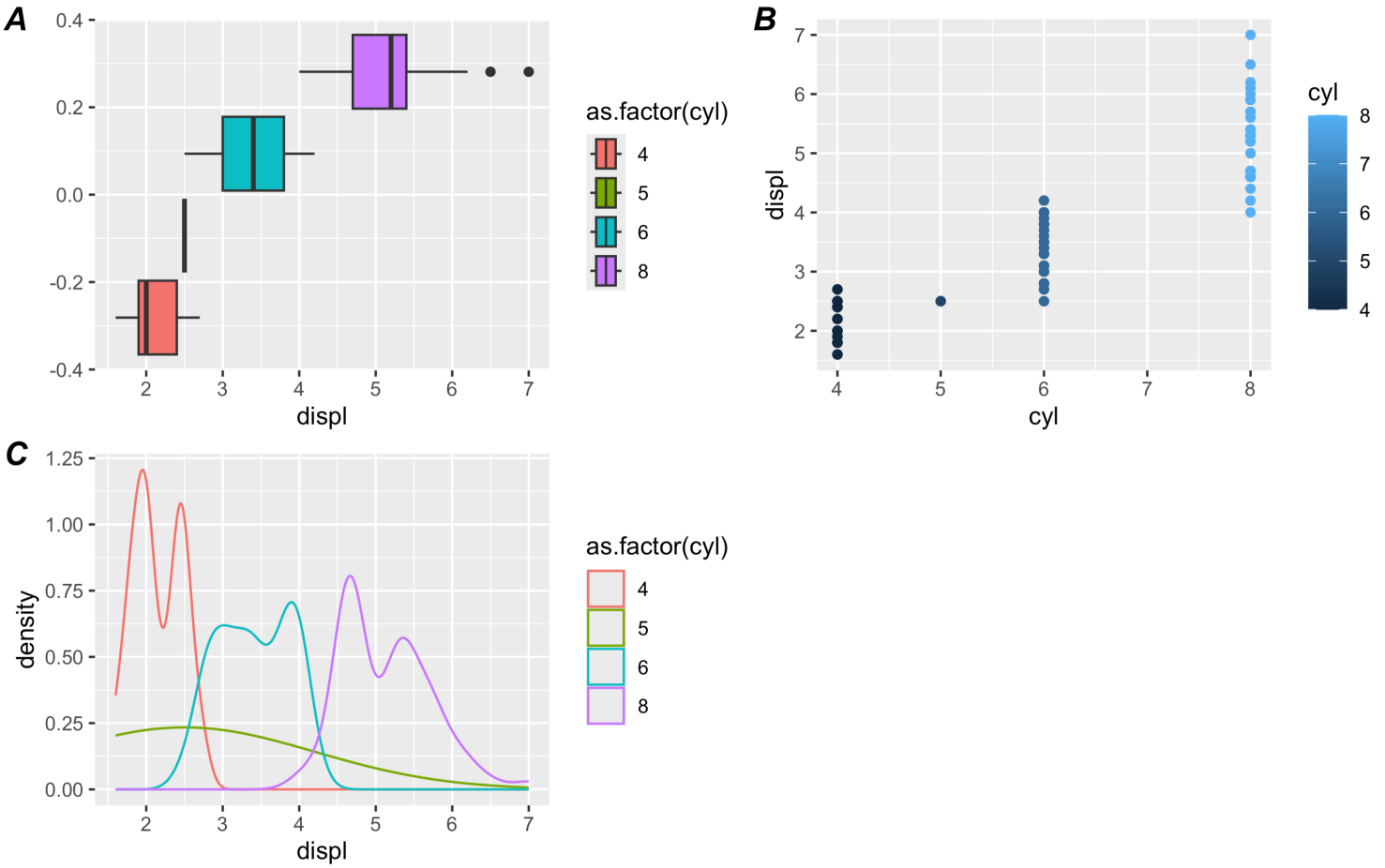
1 Matching 6 points Question

For each of the graphs shown [labeled A-C], match it to the correct code [labeled 1-6] listed below. There are more codes than graphs but there's only one correct code for each graph.




```
1) ggplot() + geom_point(mpg, mapping = aes(cyl, displ), col = "red") + geom_line()
2) ggplot(mpg, aes(displ, fill = as.factor(cyl))) + geom_boxplot()
3) ggplot(mpg, aes(displ, fill = as.factor(cyl))) +   geom_density()
4) ggplot(mpg, mapping = aes(cyl, displ, col = cyl)) + geom_point()
5) ggplot(mpg, aes(displ, col = as.factor(cyl))) +   geom_density()
6) ggplot(mpg, mapping = aes(cyl, displ), col = "red") + geom_line() + geom_point()
```

A	2
B	4
C	5

Possible answers

2 4 5 1 3 6

2 Matching 6 points Question

Identify the correct `geom_` functions that will best to create the plots mentioned below:

A plot that displays the distribution of a numeric variable along with it's median and outliers.	geom_boxplot()
A plot that displays the frequency distribution of a numeric variable using bins.	geom_histogram()
A plot that displays the distribution of categorical variables using bars.	geom_bars()

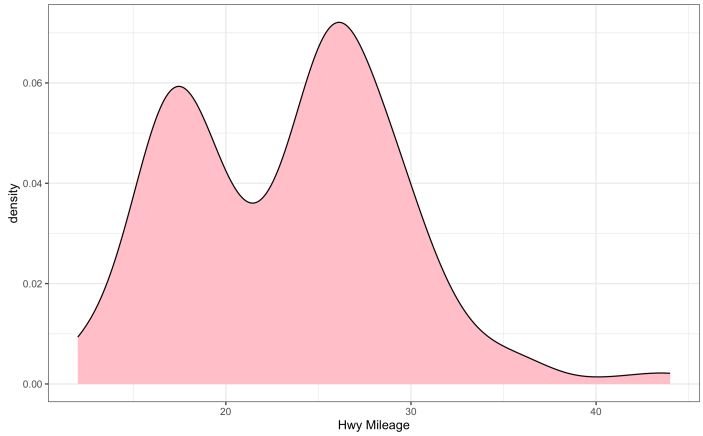
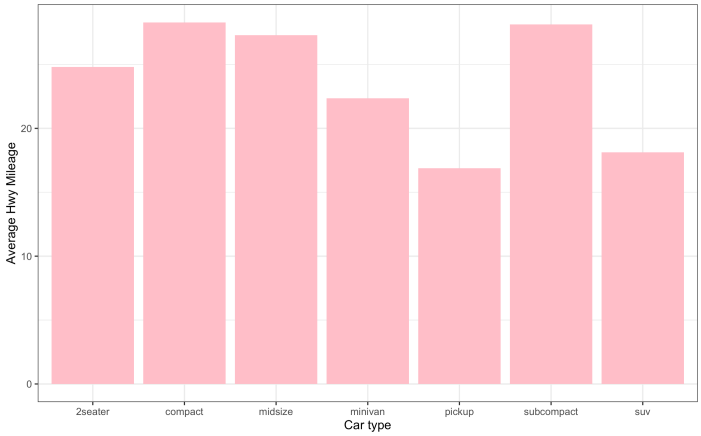
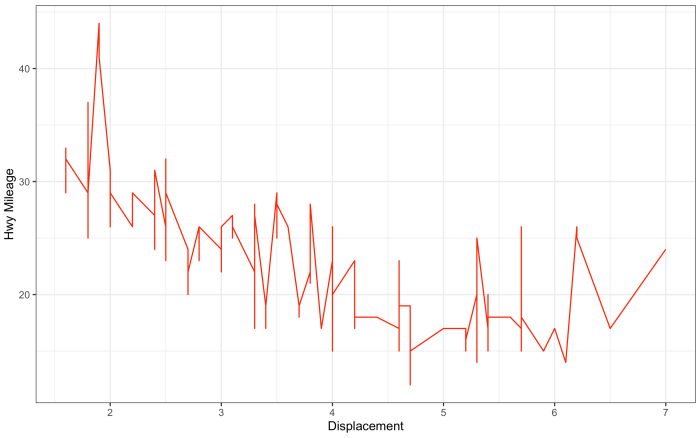
Possible answers

geom_boxplot() geom_histogram() geom_bars() geom_col()

geom_smooth() geom_density()

3 Multiple Answer 6 points Question

Pick the best `geom_` function that should be used to make the graphs below:

Graphs	`geom_` function
a) 	[density]
b) 	[column]
c) 	[line]

- [x] geom_density()
- [x] geom_col()
- [x] geom_line()

4 Fill in the Blank 5 points Question

We wish to understand which type of ships (saved in '*type*' column) in the 'ships' dataset, had more incidents (saved in '*incidents*' column) on average in each period of time (saved in '*period*' column). Fill in the blanks below with *dplyr* function (from the dropdown menu) to get a working code that gets the table shown below which will list the average number of incidents for each period and each type of ship.

```
# A tibble: 10 × 3
# Groups:   period [2]
   period    type  avg_incidents
   <chr>     <chr>         <dbl>
 1 1960-1974 A              2.25
 2 1960-1974 B             27.2
 3 1960-1974 C              1.75
 4 1960-1974 D              0.5
```

ships %>%

group_by() (period , type)

%>%

summarize() (avg_incidents = mean (incidents))

5 Fill in the Blank 5 points Question

We wish to understand which type of ships (saved in '*type*' column) in the 'ships' dataset, had maximum number of incidents (saved in '*incidents*' column) in each period of time (saved in '*period*' column). Fill in the blanks below using the dropdown menu to get a working code that gets the table shown below which will list the maximum number of incidents for each period and each type of ship.

```
# A tibble: 10 × 5
# Groups:   type, period [10]
   type  year      period    service incidents
   <chr> <chr>     <chr>       <dbl>     <dbl>
 1 A     1970-1974 1960-1974    1512         6
 2 A     1970-1974 1975-1979    3353        18
 3 B     1965-1969 1960-1974   28609        58
 4 B     1965-1969 1975-1979   20370        53
```

ships %>%

group_by (type , period)
%>%

slice_max (incidents)

6 Fill in the Blank 4 points Question

Dataset A (with **9 columns**, one of which is called `*Name*`, and **200 rows**, 1 row for each employee working in a company) and ***dataset B*** (with **10 columns**, one of which is called `*Name*`, and **2000 rows**, 1 row for each person that ever applied to the above company) are joined using the codes in the first column of the table below.

Note: Keep in mind that the all but 10 of the company's 200 employees previously applied for a job at the company. And `*Name*` is the **only** common column between A and B.

Warning: The code might not be complete but assume that it works.

Predict the number of rows and columns on the final output by filling in the blanks in the table.

Code	Rows	Coumns
`inner_join(A, B)`	190	18
`anti_join(A, B)`	10	9
`left_join(A, B)`	200	18
`full_join(A, B)`	2010	18

7 Fill in the Blank 4 points Question

Dataset A (with **9 columns**, one of which is called `*Name*`, and **150 rows**, 1 row for each employee working in a company) and ***dataset B*** (with **10 columns**, one of which is called `*Name*`, and **1000 rows**, 1 row for each person that ever applied to the above company) are joined using the codes in the first column of the table below.

Note: Keep in mind that the all but 20 of the company's 150 employees previously applied for a job at the company. And `*Name*` is the **only** common column between A and B.

Warning: The code might not be complete but assume that it works.

Predict the number of rows and columns on the final output by filling in the blanks in the table.

Code	Rows	Coumns
`right_join(A, B)`	1000	18
`semi_join(A, B)`	130	9
`anti_join(A, B)`	20	9
`full_join(A, B)`	1020	18

8 Multiple Answer 4 points Question

Consider the `movie_rating` dataset, which contains movie ratings across different platforms, as shown below.

	movie	netflix	hulu	prime
	<chr>	<dbl>	<dbl>	<dbl>
1	The Last Summer	8.2	7.8	7.5
2	Hidden Truth	7.5	8	7.2
3	Ocean Deep	8.8	8.4	8.1
4	The Bump	8.5	8.1	7.9
5	Ari's Last Song	7.8	8.3	7.6

What statements are true about the output of the code below? **SELECT ALL THAT APPLIES**.

```
movie_rating %>% pivot_longer(netflix:prime)
```

- [ ] The output dataset will have more columns than the `movie_rating` dataset
- [x] The output dataset will have less columns than the `movie_rating` dataset
- [x] The output dataset will have a column called `value`
- [x] The output dataset will have one numeric column.

9 Multiple Answer 4 points Question

Consider the `movie_rating` dataset, which contains movie ratings across different platforms, as shown below.

	movie	netflix	hulu	prime
	<chr>	<dbl>	<dbl>	<dbl>
1	The Last Summer	8.2	7.8	7.5
2	Hidden Truth	7.5	8	7.2
3	Ocean Deep	8.8	8.4	8.1
4	The Bump	8.5	8.1	7.9
5	Ari's Last Song	7.8	8.3	7.6

Which of the following statements are true regarding what we could do with this dataset? **SELECT ALL THAT APPLIES**.

- [x] This dataset is a good example of a wide dataset since it has one row for each movie
- [ ] This dataset is a good example of a long dataset since it has one row for each movie
- [x] To create the long version of this dataset using `pivot_longer()`
- [ ] To create the wide version of this dataset using `pivot_wider()`

10 Multiple Choice 4 points Question

Consider the `movie_rating` dataset, which contains movie ratings across different platforms, as shown below.

	movie	netflix	hulu	prime
	<chr>	<dbl>	<dbl>	<dbl>
1	The Last Summer	8.2	7.8	7.5
2	Hidden Truth	7.5	8	7.2
3	Ocean Deep	8.8	8.4	8.1
4	The Bump	8.5	8.1	7.9
5	Ari's Last Song	7.8	8.3	7.6

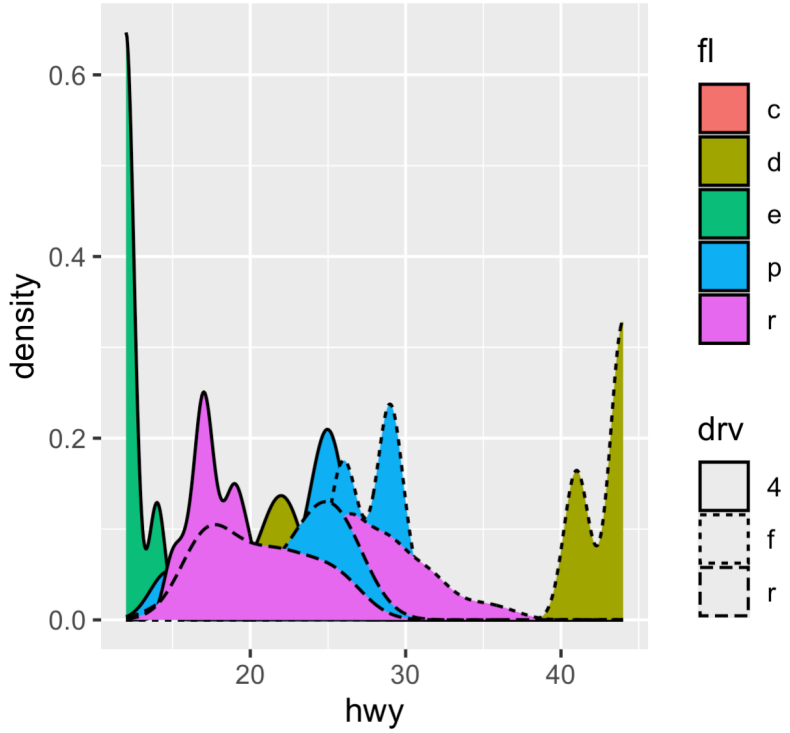
Which of the following lines of code will correctly pivot the dataset so that each movie has three rows, with the platform names stored in a column called `platform` and the corresponding ratings stored in a column called `rating`? **SELECT ONE**

- movie_rating %>% pivot_wider(netflix:prime, names_to = "platform", values_to = "rating")
- movie_rating %>% pivot_longer(netflix:prime, names_to = "platform", values_to = "rating")
- movie_rating %>% pivot_long(netflix:prime, names_from = "platform", values_from = "rating")
- movie_rating %>% pivot_wider(netflix-prime, names_from = "platform", values_from = "rating")

11 Essay 5 points Question

This is a short answer question, so write your answer in the box below.

A friend of yours created the plot shown below using the `mpg` dataset. You do not have access to the code, but you want to identify the components that were used to create the plot.



Assume that the plot was created using functions from the `ggplot2` package and that all aesthetics shown in the plot correspond to column names in the `mpg` dataset.

Your task: Identify the components that you think were used to create this plot. List the relevant **functions, arguments, and aesthetics** (for example, the variables mapped to the dataset, x-axis, y-axis, color, geoms, etc.).

You do not need to write any R code. Simply provide a list of the components that you think were used to create the graph.

The actual code is

ggplot(mpg, aes(hwy)) + geom_density(aes(fill = fl, linetype = drv))

12 True or False 2 points Question

State TRUE/FALSE.

A character column in a dataset can be sorted in `z-a` order by placing a negative sign (-) in front of the column name inside the `arrange()` function.

- ○ True
- ● False

13 True or False 2 points Question

State TRUE/FALSE.

The `arrange()` function sorts a column ascendingly by default.

- ○ True
- ○ False

14 True or False 2 points Question

State TRUE/FALSE.

`summarize()` and `summarise()` are both valid names for the same `*dplyr*` function.

- ● True
- ○ False

15 True or False 2 points Question

State TRUE/FALSE.

`filter()` can only use logical statements dealing with one column at a time.

- ○ True
- ● False

16 True or False 2 points Question

State TRUE/FALSE.

`filter()` is primarily used to select columns from a dataset using logical statements.

- ○ True
- ● False

17 True or False 2 points Question

State TRUE/FALSE.

Pipe operator (`%>%` or `|>`) can be used to put dataset B in as the second dataset in a join function.

- ○ True
- ● False

18 True or False 2 points Question

State TRUE/FALSE.

`case_when()` uses the `~` symbol to separate a the logical statement from the label for the group.

- ● True
- ○ False

19 True or False 2 points Question

State TRUE/FALSE.

Non-summary functions like `log()` can be used inside `summarize()` without causing a warning or error.

- ○ True
- ● False

20 True or False 2 points Question

State TRUE/FALSE.

Constant aesthetics such as `color = "red"` should be applied inside the `aes()` function.

- ○ True
- ● False

21 True or False 2 points Question

State TRUE/FALSE.

`ggplot()` is not absolutely necessary to build a plot using the ggplot2 package.

- ○ True
- ● False

22 True or False 2 points Question

State TRUE/FALSE.

`case_when()` is a function that can be used without the `mutate()` function.

- ○ True
- ● False

23 True or False 2 points Question

State TRUE/FALSE.

Mappings inside `aes()` are generally evaluated using variables from the **data frame**.

- (●) True
- ( ) False

24 True or False 2 points Question

State TRUE/FALSE.

The default first argument of the `ggplot()` function is the dataset but has a default value of `NULL`.

- (●) True
- ( ) False

25 Multiple Answer 6 points Question

Choose all the line of codes that will either **create an error** or **will not create** the intended output. **SELECT ALL THAT APPLIES**.

- [x] Intended Output: mean of a vector `x` with NAs in it.

  Code: mean(x, na.rm = true)

- [x] Intended output: Create a bar graph showing the counts of type of cancers in 100 cancer patients, with the outline color of the bars determined by a column called `type`.

  Code: ggplot(cancer_patient, aes(x = type)) + geom_bar(color = type)

- [ ] Intended output: Create a categorical column called `min_att` in a dataset that is `Yes` when the value of a column called `attendance` is atleast 70% and `No` when the value of the `attendance` column is below 70%.

  Code: dataset %>% mutate(min_att = case_when(attendance >= 70 ~ "Yes",

  attendance < 70 ~ "No"))

- [x] Intended output: Filter the rows that contain the numeric value 53711 in the column `zipcode` in a dataset

  Code: dataset %>% filter(zipcode = "53711")

- [x] Intended Output: Keep only the columns `name` and `age` from dataset called `df`.

  Code: df %>% relocate(name, age)

- [ ] Intended Output: Select observations/rows of a dataset `df` where the value of column `age` is between 18 and 30, inclusive.

  Code: df %>% filter(age >= 18 & age <= 30)